

Loc Nguyen

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PROFESSIONAL SUMMARY

Data analytics graduate with First Class Honours from Monash University and hands-on industry experience delivering analytics, reporting, and machine learning solutions for real-world applications. Proficient in SQL, Python, statistical analysis, dashboard development, and data visualisation, with experience turning complex datasets into actionable insights for operational and business decision-making. Strong track record of analysing trends, communicating findings clearly, and supporting scalable data-driven solutions.

EXPERIENCE

ONETASK

Data Science Intern

Melbourne, VIC

March 2023 - April 2026

- Reviewed **3,500** traffic enforcement images per day to identify phone-use and seatbelt violations, producing true-positive and false-negative statistics to support model evaluation and operational reporting.
- Conducted monthly data quality audits using **Excel VLOOKUP** to validate adjudicated outcomes against verified records, reducing the misclassification rate by **20%** and improving the reliability of labelled data used for reporting and AI evaluation.
- Identified image quality issues such as blur and poor lighting and escalated them to engineers, enabling faster camera recalibration and reducing potential data loss.
- Created **Power BI** visualisations of offence trends by highway and camera location, helping data teams identify recurring hotspots (**≈ 300** daily offenses) and communicate high-volume patterns more clearly to stakeholders.

CONSTELLATION TECHNOLOGIES LIMITED

Data Science Intern

Melbourne, VIC

July 2025 - November 2025

- Developed and benchmarked **YOLOv4, YOLO-NAS, and RF-DETR** computer vision models for phone-use and seatbelt non-compliance detection on **Linux** systems, enabling informed model selection for future production deployments.
- Enhanced detection models with the **Red-Fox Optimisation (RFO) algorithm**, achieving **94% mean average precision (mAP)** for **seatbelt** detection and **82% mAP** for **phone** detection, improving the accuracy and reliability of real-world traffic violation detection systems.
- Designed interactive visualisations to compare **YOLOv4, YOLO-NAS, and RF-DETR** training and testing performance across experiment settings such as worker count, model layers, and learning rate, improving communication of model trade-offs with operational managers and stakeholders.
- Built an **Apache Airflow** pipeline to automate training, testing, inference, and visualisation workflows, reducing manual intervention by **30%** and improving the speed and consistency of analytical reporting.

ONETASK

Data Science Intern

Melbourne, VIC

July 2024 - November 2024

- Developed a **BERT-based NLP question-answering system** to extract expiry dates and net weights from food product data, achieving **95% extraction accuracy** and improving automated data extraction efficiency.
- Built an **Isolation Forest** anomaly detection model to identify abnormal temperature patterns, achieving **99% accuracy** and supporting earlier detection of quality risks in agricultural supply chains.
- Applied **Random Forest feature importance analysis** to evaluate drivers of temperature fluctuations, improving model interpretability and identifying the most influential variables for modelling and decision-making.

- Built interactive **Power BI** reports to visualise temperature trends and the relative impact of operational variables such as storage-door opening frequency or water-pipe pressure, improving stakeholder understanding of the factors affecting product-storage conditions.

EDUCATION

Monash University

Bachelor of Applied Data Analytics Advanced (Honours)

Grade: First Class Honours (H1)

Honours Thesis: Application of Detection Transformers in Live Traffic Monitoring Systems

Dean's List 2023

Clayton, Australia

December 2025

PROJECTS

EUROPEAN BANK CHURN ANALYSIS

- Built SQL-based RFM segmentation logic and **Power BI** dashboards to analyse customer behaviour and churn patterns, improving visibility into retention risks and segment performance.
- Managed experiment tracking with **MLflow** and performed hyperparameter optimisation with **Optuna**, improving model performance and selecting the best **XGBoost** model with **72% overall accuracy**.
- Presented churn insights through a **Streamlit** interface and stakeholder-friendly visuals, making model outputs and customer risk patterns more accessible for business users.

ROSSMANN FORECASTING PROJECT

- Led a five-member team through weekly progress reviews and issue resolution across data transformation and modelling tasks, ensuring timely delivery and high-quality project outcomes.
- Performed data preprocessing and feature engineering in **Python** (Pandas), improving model input quality by handling missing values and transforming variables for downstream analysis.
- Analysed the impact of customer traffic and promotional activity on sales segments with a KNN model, achieving **98% classification accuracy** and supporting data-driven promotional planning.
- Built and optimised a Random Forest forecasting model, achieving 95% accuracy in 4-week sales prediction.

BUREAU OF METEOROLOGY PROJECT

- Collaborated in a seven-member team to preprocess raw meteorological data through 30-minute resampling and linear interpolation of missing values, creating a reliable foundation for time-series modelling.
- Engineered daily and monthly aggregated datasets and analysed the impact of key variables on rainfall patterns using **Matplotlib**, improving understanding of precipitation drivers.
- Compared **K-Nearest Neighbors**, **Logistic Regression**, and **Random Forest** models, selecting **Random Forest** as the best performer with **83% classification accuracy** in daily precipitation forecasting.

TECHNICAL SKILLS

Languages: SQL, Python, R

Libraries & Frameworks: Pandas, scikit-learn, Matplotlib, PyTorch, OpenCV

Tools & Platforms: Power BI, Apache Airflow, MLflow, Git, Docker, Linux (Ubuntu), Streamlit

Core Areas: Data Analysis, Data Visualisation, Predictive Modelling, Stakeholder Reporting